Indian Crop Production Analysis and Prediction Using Machine Learning

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***Abstract*— Agriculture plays an important role in the Indian economy and provides employment and livelihood to a large part of the population. Understanding crop production patterns can help identify changes in cultivated area, crop distribution, seasonal patterns, and agricultural productivity. This paper presents an analysis of Indian crop production data using data preprocessing, feature engineering, exploratory data analysis, and machine learning techniques. The dataset contains district-wise and season-wise crop production records from 1997 to 2022. Data cleaning was performed to handle missing values, duplicate records, invalid area values, and inconsistent numeric data. Additional features, including crop yield, production percentage, area percentage, logarithmic yield, and crop area category, were created to support analysis. Exploratory data analysis was conducted to examine year-wise cultivated area, crop distribution, state-wise cultivated area, seasonal patterns, and yield distribution. Seven regression approaches were implemented to estimate crop production: Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, K-Nearest Neighbors (KNN) Regression, Decision Tree Regression, Random Forest Regression, and Linear Support Vector Regression (Linear SVR). K-Means clustering was also applied to group records based on cultivated area and yield. Model performance was evaluated using Mean Absolute Error, Mean Squared Error, and the coefficient of determination. The initial Linear Regression experiments produced low R-squared values, indicating that further evaluation of the remaining models is necessary. The study demonstrates how data analysis and machine learning techniques can be applied to explore Indian agricultural production data.**

Keywords— Agriculture, Crop Production, Machine Learning, Data Analysis, Regression, Exploratory Data Analysis, K-Means Clustering.

I. Introduction

Agriculture is one of the important sectors of the Indian economy. It contributes to food security, employment, rural development, and economic growth. Crop production depends on several factors, including cultivated area, crop type, geographical location, agricultural season, and year. Analysing these factors can provide useful information about agricultural patterns and changes over time.

The availability of agricultural datasets provides opportunities to apply data science and machine learning techniques to understand crop production. Data analysis can help identify the distribution of crops across states and seasons, while machine learning models can be used to estimate production based on available agricultural information. The main purpose of this project is to analyse Indian crop production records and explore the application of machine learning algorithms for production prediction. The project uses district-wise, season-wise, and crop-wise agricultural records covering multiple years.

The major objectives of this study are:

1. To examine Indian crop production data and understand its structure.
2. To clean the dataset by handling missing, duplicate, and invalid records.
3. To create additional features for agricultural analysis.
4. To perform exploratory data analysis to identify crop and cultivated area patterns.
5. To implement and compare different machine learning regression algorithms.
6. To apply K-Means clustering to group agricultural records based on cultivated area and yield.
7. To evaluate model performance using standard regression metrics.

II. Dataset Description

The dataset used in this project is titled "District-wise, season-wise crop production statistics from 1997". It contains agricultural information collected across different Indian states, districts, crops, and seasons.

The dataset is listed on the Government of India's Open Government Data Platform (data.gov.in). For implementation, the CSV file was obtained from a publicly available GitHub repository. The official data.gov.in listing is cited as the dataset reference, while the GitHub repository is cited as the source of the CSV file used in this study.

The dataset contains the following seven original attributes:

Attribute	Description
State_Name	Name of the Indian state
District_Name	Name of the district
Crop_Year	Year of crop production

Season	Agricultural season
Crop	Name of the crop
Area	Cultivated area
Production	Recorded crop production

The original dataset contains 407,315 records and seven columns. The records cover the period from 1997 to 2022.

The dataset includes 34 states, 115 crop categories, and six agricultural seasons. These attributes provide information for analysing agricultural patterns across geographical locations and time periods.

A. Dataset Source

The dataset is listed at:

https://www.data.gov.in/resource/district-wise-season-wise-crop-production-statistics-1997

The CSV file used in the implementation was obtained from a GitHub mirror:

https://raw.githubusercontent.com/Ankush-Manhas840/crop-yield-prediction/master/data/crop_production.csv

The official data.gov.in listing is identified as the dataset source, while the GitHub repository is the source of the CSV file used in this implementation.

III. Data Prerrocessing and Feature Engineering

Data preprocessing is an important stage of machine learning because raw datasets may contain missing values, duplicate records, invalid values, and inconsistent data types. In this project, preprocessing was performed using Python and the Pandas and NumPy libraries.

A. Data Cleaning

The original dataset was inspected to identify missing values, duplicate records, and data types. The numerical attributes Crop_Year, Area, and Production were converted into numeric formats.

Duplicate records were removed to avoid repeated observations. Records with missing essential information, such as state, district, year, season, crop, or area, were removed. Records containing zero or negative cultivated area were also excluded.

Negative production values were removed. Records with missing production values were excluded from the production analysis dataset.

After preprocessing, the cleaned production analysis dataset contained 402,342 records.

B. Feature Engineering

Five additional features were created to support agricultural analysis.

1. **Yield:** Yield was calculated by dividing production by cultivated area. It represents production per unit of cultivated area.

$$\text{Yield} = \text{Production} / \text{Area}$$

2. **Production Percentage:** This feature represents the percentage contribution of an individual crop record to the total production within the same state, district, year, and season.
3. **Area Percentage:** This feature represents the percentage contribution of an individual crop record to the total cultivated area within the same state, district, year, and season.
4. **Production Per Area Log:** A logarithmic transformation was applied to yield using the natural logarithm of one plus yield. This transformation helps reduce the effect of highly skewed yield values during analysis.
5. **Crop Area Category:** Cultivated area values were divided into three categories: Small, Medium, and Large, using quantile-based categorisation.

These engineered features were used for additional analysis and understanding of the dataset. Features derived directly from production were not used as input variables for production prediction to avoid target leakage.

C. Categorical Encoding

Machine learning algorithms require numerical input values. Therefore, categorical attributes such as State_Name, District_Name, Season, and Crop were converted into numerical codes using Label Encoding.

The encoded attributes were used as input features in the regression models.

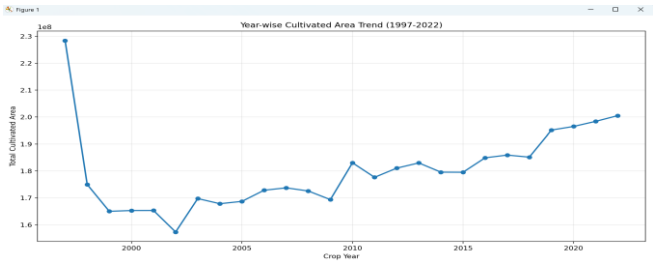
IV. Exploratory Data Analysis (EDA Process)

Exploratory Data Analysis (EDA) was performed to understand the distribution and patterns in the agricultural dataset. Python Matplotlib was used to generate graphical representations of the data.

The following analyses were conducted:

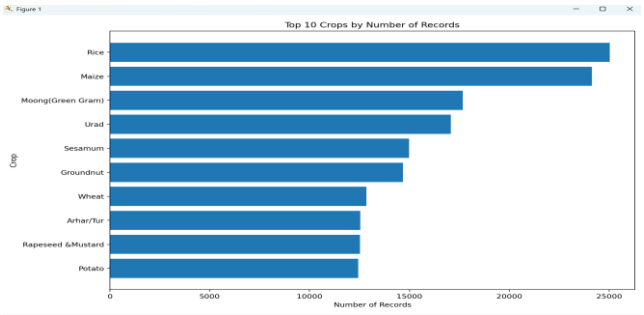
A. Year-wise Cultivated Area Trend

A line chart was created to examine changes in total cultivated area from 1997 to 2022. The chart helps observe how cultivated area changes across different years.



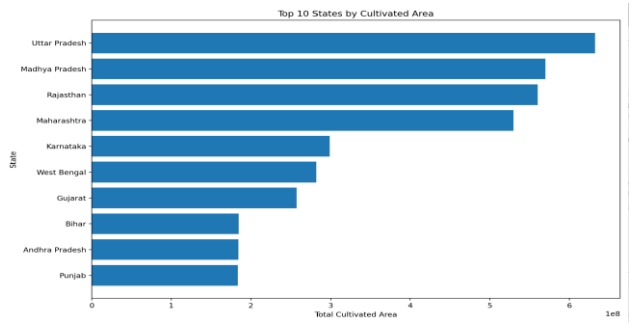
B. Top Ten Crops by Number of Records

The ten crops with the highest number of records were identified. A horizontal bar chart was used to compare their representation in the dataset.



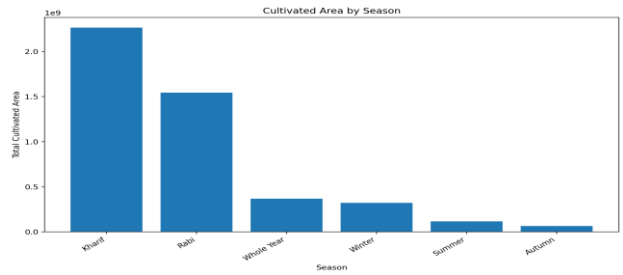
C. Top Ten States by Cultivated Area

The total cultivated area was calculated for each state. The ten states with the highest recorded cultivated area were selected and displayed using a horizontal bar chart.



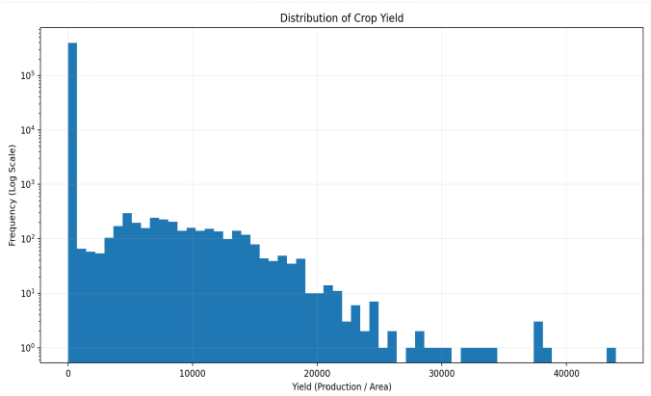
D. Cultivated Area by Season

The cultivated area was grouped according to agricultural seasons. This analysis helps compare the distribution of cultivated area across Kharif, Rabi, Whole Year, Summer, Winter, and Autumn seasons.



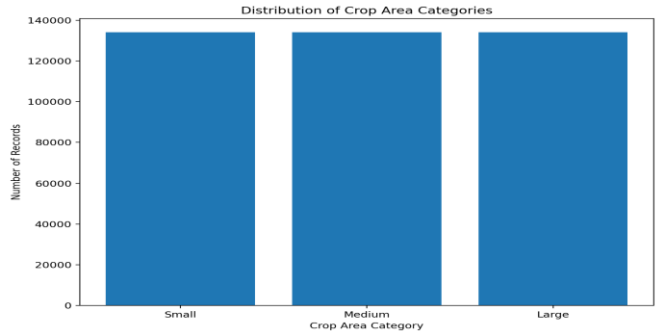
E. Distribution of Crop Yield

A histogram was generated to examine the distribution of crop yield. A logarithmic frequency scale was used to make the distribution easier to observe.



F. Crop Area Category Distribution

The records were divided into Small, Medium, and Large cultivated area categories. A bar chart was created to display the number of records in each category.



V. Machine Learning Methodology

Machine learning techniques were applied to estimate crop production using the available agricultural attributes. The target variable selected for supervised learning was Production.

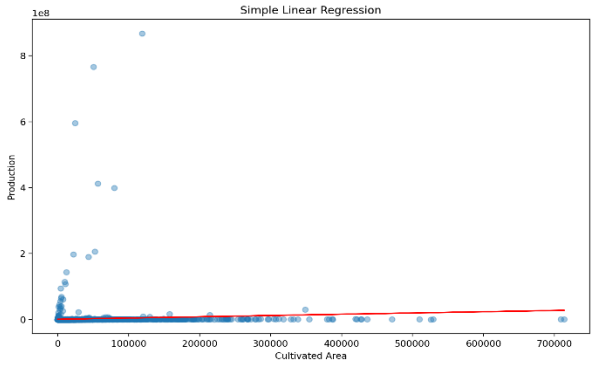
A sample of up to 30,000 records was selected from the cleaned dataset using a fixed random state to make the experiments manageable and reproducible.

The dataset was divided into training and testing subsets using an 80:20 ratio. The training data was used to fit the models, while the testing data was used to evaluate their performance.

The main input attributes included cultivated area, crop year, state, district, season, and crop. Categorical attributes were represented using their encoded numerical values.

A. Simple Linear Regression

Simple Linear Regression was implemented using cultivated area as the independent variable and production as the dependent variable. The model examines the linear relationship between cultivated area and crop production.



B. Multiple Linear Regression

Multiple Linear Regression was implemented using cultivated area, crop year, state code, district code, season code, and crop code as input variables. It considers multiple attributes while estimating crop production.

C. Polynomial Regression

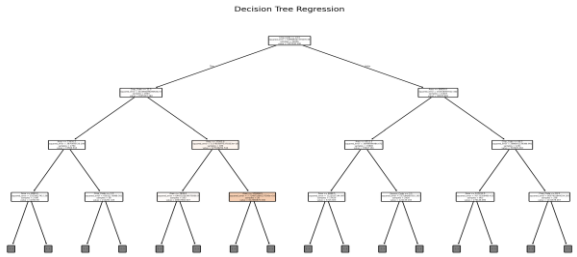
Polynomial Regression was implemented using cultivated area and crop year. Polynomial features of degree two were generated to represent possible nonlinear relationships between the input attributes and crop production.

D. K-Nearest Neighbors Regression

K-Nearest Neighbors (KNN) Regression estimates the target value using nearby observations in the feature space. In this project, five nearest neighbors were considered. StandardScaler was applied to standardise the input features because the algorithm uses distance calculations.

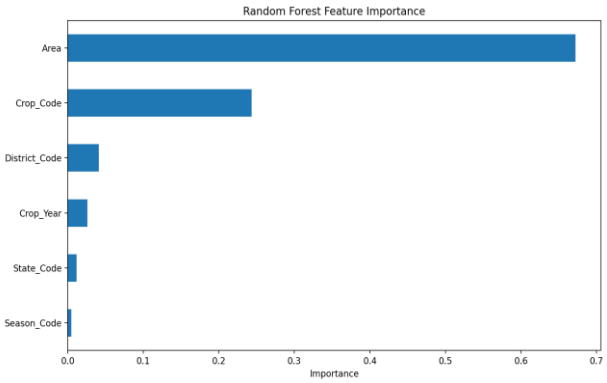
E. Decision Tree Regression

Decision Tree Regression was implemented to model relationships between agricultural input attributes and production. A maximum tree depth of five was selected to control the complexity of the model.



F. Random Forest Regression

Random Forest Regression uses multiple decision trees to generate predictions. The model was implemented using 100 estimators with a maximum depth of ten. Feature importance was also examined to understand the relative contribution of the input attributes.



G. Support Vector Regression

A linear Support Vector Regression approach was implemented using LinearSVR. It was selected as a computationally manageable support vector regression approach for the sampled dataset. StandardScaler was applied before model training.

H. K-Means Clustering\K-Means is an unsupervised learning algorithm used to group similar observations. Unlike regression models, it does not predict crop production.

In this project, cultivated area and yield were selected as clustering attributes. The features were standardised before applying K-Means with three clusters. The resulting groups were visualised using a scatter plot.

VI. Model Evaluation and Results

The regression models were evaluated using three standard performance metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and the coefficient of determination (R-squared).

A. Mean Absolute Error

MAE calculates the average absolute difference between actual and predicted values. A lower MAE indicates smaller average prediction errors.

B. Mean Squared Error

MSE calculates the average squared difference between actual and predicted values. It gives greater weight to larger prediction errors.

C. R-squared Score

R-squared measures how much of the variation in the target variable is explained by the model. A value closer to one generally indicates a better fit, while a value near zero indicates that the model explains little of the variation.

D. Initial Regression Results

The initial results obtained from Simple Linear Regression and Multiple Linear Regression are shown in Table I.

TABLE I. INITIAL REGRESSION MODEL RESULTS

Model	MAE	MSE	R-Squared
Simple Linear Regression	1,808,809.41	3.6971×10^{14}	0.00086
Multiple Linear Regression	2,834,461.05	3.6758×10^{14}	0.00661

The initial results show that both linear regression approaches produced low R-squared values. This indicates that the selected input attributes and linear relationships explain only a small proportion of the variation in crop production in these experiments.

E. Final Model Comparison

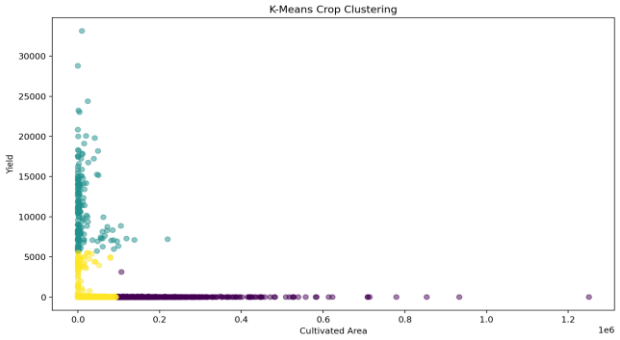
The remaining regression models must be evaluated using their actual execution results. Their metrics should be entered in Table II after the experiments are completed.

Model	MAE	MSE	R-squared
Simple Linear Regression	1,808,809.41	369,707,156,027,505.10	0.00086
Multiple Linear Regression	2,834,461.05	367,581,667,572,695.10	0.00661
Polynomial Regression	1812220.19	369800924401542.5	0.00060
KNN Regression	776620.68	176274655021060.6	0.52361
Decision Tree Regression	475436.20	111148209351537.83	0.69962
Random Forest Regression	303470.37	491439461654105410.23	0.86718
Linear SVR	844274.52	370727210315554.6	-0.00189

The final model should be selected after comparing the actual evaluation results. A higher R-squared score and lower error values generally indicate better performance on the test dataset. The computational cost and limitations of each model should also be considered.

F. K-Means Clustering Results

K-Means clustering was applied using cultivated area and yield. Three clusters were generated to group agricultural records based on similarities in these two attributes.



VII. DISCUSSION

The experimental results show differences in the performance of the seven regression approaches. Simple Linear Regression and Multiple Linear Regression produced low R-squared scores, indicating that the selected linear relationships explain only a small proportion of the variation in crop production. The performance of the other regression models was also evaluated using MAE, MSE, and R-squared to compare their prediction errors and explanatory power. These results demonstrate that different machine learning algorithms capture relationships in agricultural data to different extents.

The EDA stage provides an understanding of cultivated area patterns across years, crops, states, and seasons. The feature engineering stage creates additional attributes that help examine crop yield and the contribution of individual records within their respective groups.

The initial regression results indicate that predicting crop production is not a simple linear problem using the selected features. Crop production varies across different crops, states, districts, and agricultural seasons. The numerical encoding of categorical variables may also limit the ability of some models to represent relationships between categories correctly.

The remaining regression model results provide a basis for

comparing different approaches. K-Means clustering adds an unsupervised learning component by grouping records based on cultivated area and yield.

The study is limited by the available dataset attributes. Factors such as rainfall, temperature, soil quality, irrigation, fertilizer use, and other environmental conditions are not included in the selected model inputs. These factors may influence agricultural production but are not directly represented in the dataset used in this project.

VIII. CONCLUSION AND FUTURE SCOPE

This project presents an analysis of Indian crop production data using data preprocessing, feature engineering, exploratory data analysis, and machine learning techniques. Seven regression approaches, namely Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, K-Nearest Neighbors Regression, Decision Tree Regression, Random Forest Regression, and Linear Support Vector Regression, were implemented to estimate crop production. Additionally, K-Means clustering was applied to group agricultural records based on cultivated area and yield.

The initial linear regression experiments produced low R-squared scores, showing that the selected features and linear approaches have limited explanatory power for the sampled data. The comparison of the remaining models is necessary to identify the most suitable approach among the implemented algorithms.

The project provides practical experience in handling real-world agricultural data, performing data analysis, implementing machine learning algorithms, and evaluating model performance.

Future improvements may include:

1. Including rainfall, temperature, soil quality, and irrigation information.
2. Applying more suitable categorical encoding techniques.
3. Exploring additional feature selection and transformation methods.
4. Evaluating models using cross-validation and additional performance metrics.
5. Developing a user interface to allow users to explore agricultural data and obtain model predictions.
6. Studying crop-specific and region-specific production patterns.

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